Convergence Rates of First-Order Methods: Proofs and Numerical Experiments

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Abstract

This report studies the convergence of first-order optimization methods (gradient descent, Nesterov's accelerated gradient and stochastic gradient descent) for smooth functions. For each method we prove the classical convergence rate under convexity, strong convexity or the Polyak–Łojasiewicz condition, and we compare these theoretical bounds with numerical experiments. *[Draft: to be completed.]*

1 Introduction

[To be written: motivation, question studied, contributions, outline.]

2 Setting and assumptions

We consider the unconstrained problem

$$\min_{x \in \mathbb{R}^d} f(x),$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable, and we denote by $f^* = \inf f$ and by x^* a minimizer when one exists.

Definition 1 (L-smoothness)**.** Let $L > 0$. A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is *L-smooth* if its gradient is L-Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^d.$$

[To add: convexity, μ-strong convexity, condition number $\kappa = L/\mu$, Polyak–Łojasiewicz condition.]

3 Preliminary results

Lemma 1 (Descent lemma)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. Then for all $x, y \in \mathbb{R}^d$,*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2 .$$

Proof. Since ∇f is L-Lipschitz, it is continuous, hence f is of class C^1. Fix $x, y \in \mathbb{R}^d$ and set $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. The function φ is C^1 with $\varphi'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$, so by the fundamental theorem of calculus,

$$f(y) - f(x) = \varphi(1) - \varphi(0) = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle \, dt.$$

Since $\int_0^1 \langle \nabla f(x), y-x\rangle \, dt = \langle \nabla f(x), y-x\rangle$, we obtain

$$f(y) - f(x) - \langle \nabla f(x), y-x\rangle = \int_0^1 \langle \nabla f(x+t(y-x)) - \nabla f(x), y-x\rangle \, dt.$$

By the Cauchy–Schwarz inequality, then L-smoothness,

$$\begin{aligned}
\int_0^1 \langle \nabla f(x+t(y-x)) - \nabla f(x), y-x\rangle \, dt &\le \int_0^1 \|\nabla f(x+t(y-x)) - \nabla f(x)\| \, \|y-x\| \, dt \\
&\le \int_0^1 L \, \|t(y-x)\| \, \|y-x\| \, dt \\
&= L \, \|y-x\|^2 \int_0^1 t \, dt \qquad \text{(since } t \ge 0\text{)} \\
&= \frac{L}{2} \, \|y-x\|^2 ,
\end{aligned}$$

which gives the result. □

Remark 1. Convexity is not used in the proof: the descent lemma holds for every L-smooth function. This is what allows the analysis of gradient descent for non-convex functions satisfying the Polyak–Łojasiewicz condition (Section 4.4).

Remark 2. Since the Cauchy–Schwarz inequality bounds the absolute value of the integrand, the same argument gives the two-sided bound

$$|f(y) - f(x) - \langle \nabla f(x), y-x\rangle| \le \frac{L}{2} \, \|y-x\|^2 \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Corollary 1 (Sufficient decrease)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. For every $x \in \mathbb{R}^d$, the gradient step $x^+ = x - \frac{1}{L}\nabla f(x)$ satisfies*

$$f(x^+) \le f(x) - \frac{1}{2L} \, \|\nabla f(x)\|^2 .$$

Proof. Applying Lemma 1 with $y = x^+$, so that $x^+ - x = -\frac{1}{L}\nabla f(x)$, we get

$$\begin{aligned}
f(x^+) &\le f(x) + \left\langle \nabla f(x), -\tfrac{1}{L}\nabla f(x) \right\rangle + \frac{L}{2} \left\| -\tfrac{1}{L}\nabla f(x) \right\|^2 \\
&= f(x) - \frac{1}{L} \, \|\nabla f(x)\|^2 + \frac{L}{2L^2} \, \|\nabla f(x)\|^2 \\
&= f(x) - \frac{1}{2L} \, \|\nabla f(x)\|^2 .
\end{aligned}$$

□

Remark 3. In particular, each gradient step with step size $1/L$ decreases f, strictly as long as $\nabla f(x) \neq 0$. As for Lemma 1, convexity is not needed. Iterating the inequality along the gradient descent sequence $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$ gives the starting point of all the convergence proofs of Section 4.

4 Gradient descent

Throughout this section, $(x_k)_{k\ge 0}$ denotes the gradient descent sequence with step size $1/L$:

$$x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k), \qquad k \ge 0.$$

4.1 The quadratic case: how tight is the sufficient decrease?

Let $A \in \mathcal{S}_d(\mathbb{R})$ be symmetric positive definite, with smallest eigenvalue $\mu > 0$ and largest eigenvalue L, and let

$$f(x) = \frac{1}{2} x^\top A x, \qquad \nabla f(x) = Ax.$$

Then f is L-smooth and μ-strongly convex, with $x^* = 0$ and $f^* = 0$. To measure how much of the guaranteed decrease of Corollary 1 is actually achieved, we define, whenever $\nabla f(x_k) \neq 0$,

$$r_k = \frac{f(x_k) - f(x_{k+1})}{\frac{1}{2L} \|\nabla f(x_k)\|^2}.$$

Corollary 1 states exactly that $r_k \geq 1$.

Proposition 1. *For every $k \geq 0$ such that $x_k \neq 0$,*

$$r_k = 2 - \frac{\langle g_k, A g_k \rangle}{L \|g_k\|^2} \qquad \text{where } g_k = \nabla f(x_k) = Ax_k,$$

and consequently

$$1 \leq r_k \leq 2 - \frac{\mu}{L} = 2 - \frac{1}{\kappa}.$$

Proof. Since A is invertible, $x_k \neq 0$ implies $g_k \neq 0$, so r_k is well defined. Write $x = x_k$ and $g = g_k = Ax$, so that $x_{k+1} = x - \frac{1}{L} g$.

Exact expression. By definition of r_k and of f,

$$r_k = \frac{2L}{\|g\|^2} \Big(\tfrac{1}{2} x^\top A x - \tfrac{1}{2} (x - \tfrac{1}{L} g)^\top A (x - \tfrac{1}{L} g) \Big) = \frac{L}{\|g\|^2} \Big(\langle x, Ax \rangle - \Big\langle x - \tfrac{1}{L} g, Ax - \tfrac{1}{L} Ag \Big\rangle \Big).$$

Expanding the second inner product and using $Ax = g$,

$$\Big\langle x - \tfrac{1}{L} g, g - \tfrac{1}{L} Ag \Big\rangle = \langle x, g \rangle - \frac{1}{L} \langle x, Ag \rangle - \frac{1}{L} \langle g, g \rangle + \frac{1}{L^2} \langle g, Ag \rangle .$$

Since A is symmetric, $\langle x, Ag \rangle = \langle Ax, g \rangle = \|g\|^2$, and $\langle x, Ax \rangle = \langle x, g \rangle$. Therefore

$$r_k = \frac{L}{\|g\|^2} \Big(\frac{2}{L} \|g\|^2 - \frac{1}{L^2} \langle g, Ag \rangle \Big) = 2 - \frac{\langle g, Ag \rangle}{L \|g\|^2}.$$

Bounds. By the spectral theorem, A is diagonalizable in an orthonormal basis $(e_1, \ldots, e_d)$ of eigenvectors, with eigenvalues $\lambda_1, \ldots, \lambda_d \in [\mu, L]$. Writing $g = \sum_{i=1}^d \langle g, e_i \rangle e_i$, we get $Ag = \sum_{i=1}^d \lambda_i \langle g, e_i \rangle e_i$, hence

$$\langle g, Ag \rangle = \sum_{i=1}^d \lambda_i \langle g, e_i \rangle^2 .$$

Since $\mu \leq \lambda_i \leq L$ and $\sum_{i=1}^d \langle g, e_i \rangle^2 = \|g\|^2$ (Parseval's identity in an orthonormal basis),

$$\mu \|g\|^2 \leq \langle g, Ag \rangle \leq L \|g\|^2 , \qquad \text{i.e.} \qquad \frac{\langle g, Ag \rangle}{L \|g\|^2} \in \Big[\frac{\mu}{L}, 1 \Big].$$

Plugging this into the exact expression gives $1 \leq r_k \leq 2 - \frac{\mu}{L}$. □

Remark 4. The quantity $\langle g, Ag \rangle / \|g\|^2$ is the Rayleigh quotient of A at g. The lower bound $r_k = 1$ is attained when g_k is an eigenvector for L: the sufficient decrease inequality is then tight. The upper bound $2 - 1/\kappa$ is attained when g_k is an eigenvector for μ. Along gradient descent, the components of x_k along the eigenvector e_i are multiplied by $(1 - \lambda_i / L)$ at each step, so the slowest direction (eigenvalue μ) dominates for large k, and one expects $r_k \to 2 - 1/\kappa$. This is what we observe numerically (Section 7).

4.2 Convex case

[R1: $f(x_k) - f^* \leq \dfrac{L \left\| x_0 - x^* \right\|^2}{2k}$*.]*

4.3 Strongly convex case

[R2: $\left\| x_k - x^* \right\|^2 \leq (1 - 1/\kappa)^k \left\| x_0 - x^* \right\|^2$*.]*

4.4 Polyak–Łojasiewicz condition

[R4: linear rate without convexity.]

5 Nesterov's accelerated gradient

[R3: $O(1/k^2)$ *via a Lyapunov function;* $(1 - 1/\sqrt{\kappa})^k$ *in the strongly convex case; lower bound; ODE interpretation.]*

6 Stochastic gradient descent

[R5: constant step size and noise floor; decreasing step size and $O(1/k)$ *rate.]*

7 Numerical experiments

[Quadratics with controlled κ*; logistic regression; non-convex PL example; SGD on least squares; comparison with Adam.]*

8 Discussion

[Where the bounds are tight, where they are pessimistic, and why.]

References

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